

Soft Computing

TCS 821

Unit I: Fundamentals of ANN

1. Define Artificial Neural Network (ANN) and explain its biological inspiration.
2. Describe ANN terminology such as neuron, weight, bias, and activation function.
3. Explain the architecture of an Artificial Neural Network.
4. What are activation functions? Explain different types.
5. Describe the Perceptron learning rule.
6. Explain the Delta learning rule and its significance.
7. Compare biological neural networks and artificial neural networks.
8. Explain the structure and working of a Biological Neural Network. Compare it with ANN, highlighting the key differences.
9. Describe the building blocks of Artificial Neural Networks (ANN). Explain its architecture, weights, and activation functions.
10. Explain McCulloch–Pitts Neuron Model and Hebbian Learning Rule in detail.

Unit II: Models of ANN

1. Explain the architecture of a single-layer perceptron.
2. Describe the algorithm of a perceptron model.
3. What are feedback networks? Explain Hopfield Network.
4. Describe Bidirectional Associative Memory (BAM).
5. What are feedforward networks? Explain their characteristics.
6. Explain the Backpropagation Network (BPN) with algorithm.
7. What is Radial Basis Function Network (RBFN)? Explain its working.
8. Describe Self-Organizing Maps (SOM).
9. Explain Learning Vector Quantization (LVQ) and compare it with SOM.
10. Describe Hopfield Network and Bidirectional Associative Memory (BAM)

Unit III: Fuzzy Sets and Relations

1. Define fuzzy sets and explain their properties.
2. Explain operations on fuzzy sets (union, intersection, complement).
3. What are fuzzy relations? Explain with examples.
4. Define cardinality of fuzzy relations.
5. Explain operations on fuzzy relations.
6. Discuss properties of fuzzy relations.
7. What is fuzzy composition? Explain its types.

8. Explain max-min composition in fuzzy relations.
9. Explain Fuzzy Relations, including cardinality, types, and properties with mathematical representation.
10. Describe Fuzzy Composition and explain its significance in fuzzy systems with examples.

Unit IV: Fuzzy Logic and Inference Systems

1. Define fuzzy variables and explain their significance.
2. What are membership functions? Explain different types.
3. Differentiate between Takagi and Mamdani fuzzy models.
4. What is fuzzification? Explain its process.
5. Describe the inference mechanism in fuzzy systems.
6. Explain the concept of rule base in fuzzy systems.
7. What is defuzzification? Explain different methods.
8. Describe the working of a fuzzy inference system.
9. Define Fuzzy Variables and explain different types of Membership Functions with diagrams.
10. Describe the Fuzzy Inference System (FIS) process, along Fuzzification, Rule Base, Inference Engine and Defuzzification

Unit V: Genetic Algorithm (GA)

1. Define Genetic Algorithm and explain its biological inspiration.
2. Explain the basic terminology used in Genetic Algorithms.
3. Describe the encoding techniques used in GA.
4. What are selection methods in GA? Explain different types.
5. Explain crossover in GA and its types.
6. What is mutation? Explain its role in GA.
7. Explain the steps involved in a Genetic Algorithm.
8. Explain the biological terminology used in Genetic Algorithms and relate it to computational concepts.
9. Describe the elements of a Genetic Algorithm with Encoding, Selection Methods, Crossover Types, Mutation, and Reinsertion Strategies
10. Explain the working of a Simple Genetic Algorithm and discuss its theoretical foundation, and Schema Theorem and Building Block Hypothesis